Parallel Compared-and-Stacked Pyramid Transformer Network for Unsupervised Hyperspectral Change Detection

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***Abstract*— Convolutional neural networks (CNNs) with good feature learning capabilities are widely used in hyperspectral image change detection (HSI-CD) tasks. However, most existing CNN-based HSI-CD methods face two inherent challenges: 1) the lack of available labeled datasets and 2) the limited receptive field that cannot capture the long-distance dependence between the spectral sequences of HSIs. In this article, we propose a parallel compared-and-stacked pyramid transformer network (PCPTNet) for unsupervised HSI-CD, which can model the context information of spectral sequences in the input multi-temporal HSI patch without real labeled data. Specifically, a superpixel-level joint decision-based training samples selection strategy is presented that fully considers the correlation between pixels to improve the reliability of training samples. Then, taking advantage of transformer in context information modeling, PCPTNet is proposed to capture sufficient difference features and stacked features with different scales for CD, which can effectively reduce missed and false detection. The multiscale features containing sufficient low-level detail information and high-level semantic features are fused hierarchically to classify changed and unchanged pixels. Extensive experiments on three real HSI datasets demonstrate that the PCPTNet outperforms other state-of-the-art HSI-CD methods in both visual and quantitative results.**

***Index Terms*— Hyperspectral image change detection (HSI-CD), joint decision, pyramid, transformer, unsupervised.**

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I. Introduction

THE surface of the earth is constantly changing under the influence of natural factors (e.g., climate change and natural disasters) and human activities (e.g., urbanization and industrialization) [1]. Change detection (CD), as a technique for identifying changes by analyzing the multiple images acquired over the same surface area of interest at different times, provides an effective way to detect these changes [2]. Remote sensing technology that can obtain remote sensing images over large-scale areas, especially inaccessible areas, has been widely used in earth observations [3], [4].

As one of the most important remote sensing data, hyperspectral images (HSIs) contain rich and fine spectral information [5], [6], [7], and show great potential in identifying changes in multi-temporal images [8], [9]. In HSI, each pixel corresponds to a spectral vector that reflects the spectral response of the object over hundreds of bands, which can provide a particularly fine and comprehensive representation of the properties of the object. In practice, some objects have only minor changes in radiability at specific spectral bands due to small differences in chemical composition. In addition, the complexity of the objects contained in the image often results in diverse changes. These minor and diverse changes are difficult to be detected in most remote sensing images, but can be detected by analyzing the fine spectral vectors of HSIs. Therefore, HSIs with extremely high spectral resolution can provide sufficient spectral information to detect minor and diverse changes, thus improving the accuracy of CD [10].

In recent years, hyperspectral image change detection (HSI-CD) as an effective earth observation remote sensing technology has been widely used in both military and civilian fields. With the continuous development of remote sensing technology and the increase of available HSI datasets, a variety of traditional HSI-CD methods have been proposed, which can be roughly summarized into four groups: 1) image algebra-based CD; 2) transformation-based CD; 3) object classification-based CD; and 4) other methods. The image algebra-based HSI-CD methods directly use the difference in image gray levels to determine the changes pixel by pixel and generate a difference map. As a widely used image algebra-based CD method, change vector analysis (CVA) [11] can make full use of continuous spectral vectors of

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HSIs and analyze the difference in amplitude and direction between corresponding spectral vectors of multi-temporal HSIs to detect changes. However, the image algebra-based HSI-CD methods only consider the spectral information of the HSIs and ignore the spatial relationship, resulting in low detection accuracy. The transformation-based methods learn a new transformed feature representation that can highlight the changed features and suppress the unchanged features from the original multi-temporal HSIs to easily determine the change and nonchange class. Typical transformation-based CD methods mainly include principal component analysis (PCA) [12] and iteratively reweighted multivariate alteration detection (IR-MAD) [13]. PCA transforms high-dimensional HSI into a low-dimensional principal component space to reduce redundant information and extract features with the most variable information for CD. IR-MAD utilizes canonical correlation analysis and adaptive weight strategies to extract precise and robust difference features of multi-temporal HSIs, facilitating the recognition of changes. However, it is clear that there is inevitably some degree of spectral distortion in the transformation process, which will affect the accuracy of CD. The object classification-based CD methods usually design a classifier to treat CD as a binary classification task. Recently, a large number of CD methods based on object classification are proposed, including support vector machines (SVMs) [14] and maximum likelihood classifier (MLC) [15]. The object classification-based detection methods have been successfully applied in practice [16]. However, it is worth noting that the accuracy of the object classification based-CD methods strongly depends on the performance of the classifier. Furthermore, some widely used techniques have also been successfully introduced into the HSI-CD tasks, such as unmixing [17], relative radiometric normalization [18], and sparse decomposition [19]. Du et al. [20] adopted linear unmixing technology to realize CD, which extracts the abundance and endmember information of each multi-temporal image data to distinguish the change. Alp et al. [21] successfully applied sparse unmixing and dictionary pruning to HSI-CD.

The aforementioned methods can only extract handcrafted shallow feature representation, which have difficulties in exploiting the crucial high-level information of HSIs. In recent years, deep learning models with good feature learning capabilities have achieved superior performance in many image processing fields [22]. The ability of deep learning to automatically obtain the characteristics of the input image is particularly suitable for CD tasks. For HSI-CD task, the key is to explore the correlation between pixels in multi-temporal HSIs. Due to the influence of atmospheric scattering and complex light scattering mechanism, the detection of changes between corresponding pixels is essentially a nonlinear process. Deep learning models can efficiently and automatically extract nonlinear features of input data that are especially suitable to fit the correlations between pixels in multi-temporal HSIs for CD. Meanwhile, deep learning models with powerful feature modeling and learning capabilities can effectively learn and interpret abstract high-dimensional features of HSIs, providing sufficient information for discriminating changes. Wang et al. [23] proposed a CD network named general end-to-end 2-D convolutional neural network (GETNET), in which 2-D convolution was adopted to solve the high-dimensional problem of HSI data and obtain sufficient feature information. Dong at al. [24] highlighted the difference information at the subpixel level to detect the changed regions of multi-temporal HSIs. Shi et al. [25] used long short-term memory (LSTM) and convolutional neural network (CNN) to model multiscale temporal–spatial–spectral features of HSI to improve detection accuracy. Wang et al. [26] proposed an end-to-end residual self-calibration network (RSCNet) to improve the detection accuracy with fewer additional parameters. Ge et al. [27] presented a multidirection and multiscale spectral–spatial residual network to investigate feature variation of HSIs, thus improving detection accuracy. Wang et al. [28] proposed an end-to-end Siamese CNN with a spectral-spatial-wise attention mechanism (SSA-SiamNet) for HSI-CD, which can adaptively refines the spectral-spatial features to increase the detection accuracy. Hu et al. [29] proposed a novel partial-Siamese united-unmixing module to makes the abundance of the unchanged pixels more similar for CD. In general, the detection accuracy of the CD methods based on deep learning is significantly higher than that of the traditional methods. However, the CNN-based HSI-CD methods still face some challenges, such as the lack of labeled datasets and limited receptive fields.

The generalization of deep learning methods generally relies on a large number of labeled training samples. However, it is time-consuming and labor-intensive to obtain high-quality labeled samples in practice. For some steep areas and objects containing confidential information, it is impossible to obtain any labeled samples. Unsupervised learning provides an effective solution to the problem of shortage of labeled datasets. However, only a few studies focus on achieving HSI-CD in an unsupervised learning manner. Li et al. [30] used two complementary model-driven methods to provide credible labels for the training of network to achieve unsupervised HSI-CD. Zhao et al. [31] designed an unsupervised sample generation strategy for the training of an efficient feature extraction network to facilitate the recognition of changed and unchanged regions in multi-temporal HSIs. Obviously, the quantity and quality of labels directly affect the detection performance. Most existing methods generate training sample labels at the pixel level which ignores the correlation between pixels and may lead to low reliability of labels. Another important challenge in the deep learning-based HSI-CD task is that the convolution operation with a fixed-size convolution kernel in CNN often has limited receptive fields and cannot fully exploit the correlation between high-dimensional spectral vectors, which leads to difficulties in identifying changed objects. Transformer based on multi-head self-attention (MHSA) mechanism has a good ability to capture long-distance dependencies between spectral sequences, which can provide sufficient context information for the discrimination of target pixels and improve the detection performance.

Inspired by these, in this article, a parallel compared-and-stacked pyramid transformer network (PCPTNet) for unsupervised HSI-CD is proposed, which uses a paral-

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lel compared-and-stacked pyramid structure to introduce transformer into unsupervised CD tasks to achieve excellent CD performance. As an unsupervised CD method, the PCPTNet can effectively get rid of the dependence on labeled data, which is conducive to the application of CD in real scenes. In PCPTNet, the transformer based on the MHSA mechanism can model the context information of the input image patch, which can fully explore the correlation between the spectral sequences internally to improve the accuracy of CD. In addition, the parallel compared-and-stacked pyramid structure composed of the compared pyramid transformer (CPT) subnetwork and the stacked pyramid transformer (SPT) subnetwork considers both image comparison operations and image stacking operations, which provides a good way to reduce missed detection and false detection. Finally, the corresponding outputs of each layer of the CPT subnetwork and SPT subnetwork are fused to realize discrimination and prediction, which effectively utilizes multiscale features to enhance the ability of the PCPTNet to distinguish objects in multi-temporal HSIs. In summary, the main contributions of our work can be summarized below.

1) As an unsupervised HSI-CD model, PCPTNet innovatively presents a joint decision strategy at the superpixel level to obtain the high-confidence labels of training samples, which fully considers the spatial–spectral correlation between pixels and overcomes the challenge of the lack of labeled datasets.
2) The pyramid transformer network is designed to obtain multiscale features of multi-temporal HSIs. For each input image patch, the transformer treats each spectral vector as an input sequence, which can effectively explore the contextual information between pixels in the image patch. As a result, the feature pyramid transformer network has the superior ability to acquire multiscale features that are conducive to the identification of objects in HSI.
3) The parallel compared-and-stacked pyramid structure is innovatively proposed to combine the advantages of the CPT subnetwork in acquiring difference features and the SPT subnetwork in preventing information loss, greatly reducing missed and false detection of CD results.

The remainder of this article is organized as follows. The Section II introduces the related work of the proposed PCPTNet. The Section III describes the proposed PCPTNet in detail. In Section IV, extensive experimental results are analyzed to verify the effect of the PCPTNet. In the end, the conclusion of this article is summarized in Section V.

II. Related Work

Transformer, as an efficient model proposed by the Google team for natural language processing (NLP) tasks [32], has attracted great attention of researchers owing to its superior performance in sequence modeling. Instead of the local convolution kernel of CNN or the sequential structure of recurrent neural network (RNN), the transformer model mainly composed of MHSA and multilayer perceptron (MLP), can fully explore the global relationship of the input sequence in parallel.

Formally, given a set of input sequences $\{u_1, \ldots, u_i, \ldots, u_m\}$, $u_i \in \mathbb{R}^{1\times s}$, we organize them into a matrix for simultaneous processing. The input sequences are mapped to a constant dimension D through a linear projection matrix, and then position embedding is added to maintain position information [33]. The obtained embedded input $\mathbf{I} \in \mathbb{R}^{m\times D}$ is then fed into the transformer backbone. In transformer, the self-attention module that uses query, key, and value matrices $\mathbf{Q} \in \mathbb{R}^{m\times d}$, $\mathbf{K} \in \mathbb{R}^{m\times d}$, and $\mathbf{V} \in \mathbb{R}^{m\times d}$ to obtain the internal correlation of feature sequences has the advantage of focusing on important information of the input sequences. Further, the MHSA composed of multiple self-attention modules can explore the relationship between sequences from multiple subspaces. This process can be formulated as

$$\begin{aligned}
\text{Att}(\mathbf{Q},\mathbf{K},\mathbf{V}) &= \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \\
\text{MHSA}(\text{LN}(\mathbf{I})) &= \text{Concat}\left(h_1, \ldots, h_j, \ldots, h_z\right)\mathbf{W}^0, \quad \text{where} \\
h_j &= \text{Att}\left(\mathbf{Q}_j, \mathbf{K}_j, \mathbf{V}_j\right) \\
&= \text{Att}\left(\text{LN}(\mathbf{I})\mathbf{W}_j^q, \text{LN}(\mathbf{I})\mathbf{W}_j^k, \text{LN}(\mathbf{I})\mathbf{W}_j^v\right) \quad (1)
\end{aligned}$$

where $\mathbf{W}_j^q \in \mathbb{R}^{D\times d}$, $\mathbf{W}_j^k \in \mathbb{R}^{D\times d}$, $\mathbf{W}_j^v \in \mathbb{R}^{D\times d}$, and $\mathbf{W}^0 \in \mathbb{R}^{zd\times D}$ are weight matrices, z is the number of self-attention modules, softmax$(\cdot)$ denotes the normalized exponential function, LN$(\cdot)$ represents the layer normalization operation, and Concat$(\cdot)$ represents the concatenation operation. Compared with single-head self-attention, MHSA can comprehensively utilize information from different self-attention modules to obtain richer features and improve network performance. Then, residual connections are introduced to improve the stability of the network. To further enhance the feature expression ability of the transformer model, the output of the residual connection of the MHSA part is fed to the MLP part composed of layer normalization [34], MLP, and residual connection. The MLP allows sufficient information interaction between feature vectors in different dimensions, which can capture sufficient nonlinear features to improve the performance of transformer model [35]. In this way, combined with the advantages of MHSA in modeling long-term sequences and MLP in extracting nonlinear features, the transformer has a strong ability to capture the context information of the input sequence and extract sufficient features of the objects in the image.

Inspired by the advantages of transformer in processing sequence data, researchers make great efforts to introduce transformer into HSI processing fields such as image classification [36] and image super-resolution [37]. Nevertheless, only a few studies focus on introducing transformer into the HSI-CD task to improve the detection accuracy. Wang et al. [38] adopted transformer to simultaneously model the spectral, spatial, and temporal sequence information of HSI to improve the detection accuracy. Ding et al. [39] used spatial encoding, temporal encoding and multihead attention mechanism of transformer to jointly extract the spatial-spectral-temporal features of HSIs and improve the discrimination ability of the network. However, most existing transformer-based HSI-CD methods rely on real labeled data

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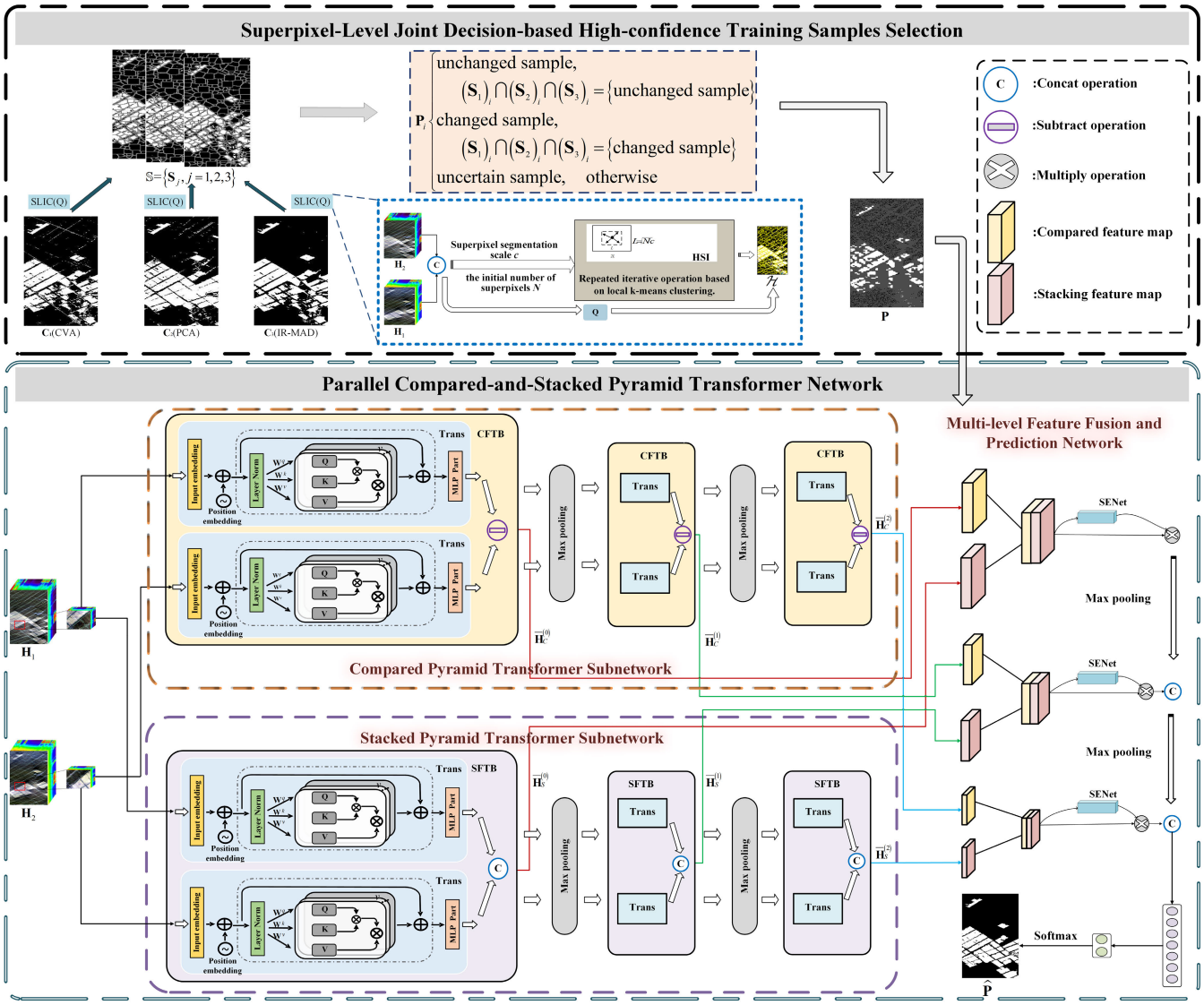


Fig. 1. Network structure of the proposed PCPTNet for unsupervised HSI-CD.

for network training. In practice, the labeling of samples is very difficult, which greatly limits the application and generalization of CD techniques. To address this problem, in this article, we propose an unsupervised HSI-CD network, in which transformer is introduced to fully mine the context information of HSI so as to improve the feature expression ability and detection accuracy of the network.

III. Proposed Method

In this section, the proposed PCPTNet for unsupervised HSI-CD is elaborated. As illuminated in Fig. 1, the overall framework of PCPTNet can be roughly divided into two parts: 1) superpixel-level joint decision-based high-confidence training samples selection and 2) PCPTNet. It is well known that the labeled training samples essential for network training are generally difficult to obtain in practice, especially for complex remote sensing data. To address this problem, a novel superpixel-level joint decision-based high-confidence training samples selection strategy is proposed in this article. Given two input multi-temporal HSIs, three traditional unsupervised CD methods (i.e., CVA, PCA, and IR-MAD) with stable detection performance are used to obtain the initial CD maps, respectively. Then, the superpixel regions of the initial CD maps are obtained according to the superpixel distribution of the multi-temporal HSIs. A joint decision voting rule is designed to screen out high-confidence pixels from each superpixel region as the training sample labels of PCPTNet. The PCPTNet is mainly composed of a CPT subnetwork, a SPT subnetwork, and a multilevel feature fusion and prediction network. The CPT subnetwork is designed to obtain the difference features of multi-temporal HSIs to facilitate discriminating differences, and the SPT subnetwork can prevent the loss of feature information to reduce missed detection. Multilevel feature fusion and prediction network merges the features of CPT subnetwork and SPT subnetwork at corresponding levels to identify the changed and unchanged pixels.

A. Superpixel-Level Joint Decision-Based High-Confidence Training Samples Selection

Deep learning-based models rely heavily on the ground truth during the network training. However, in practice, many areas

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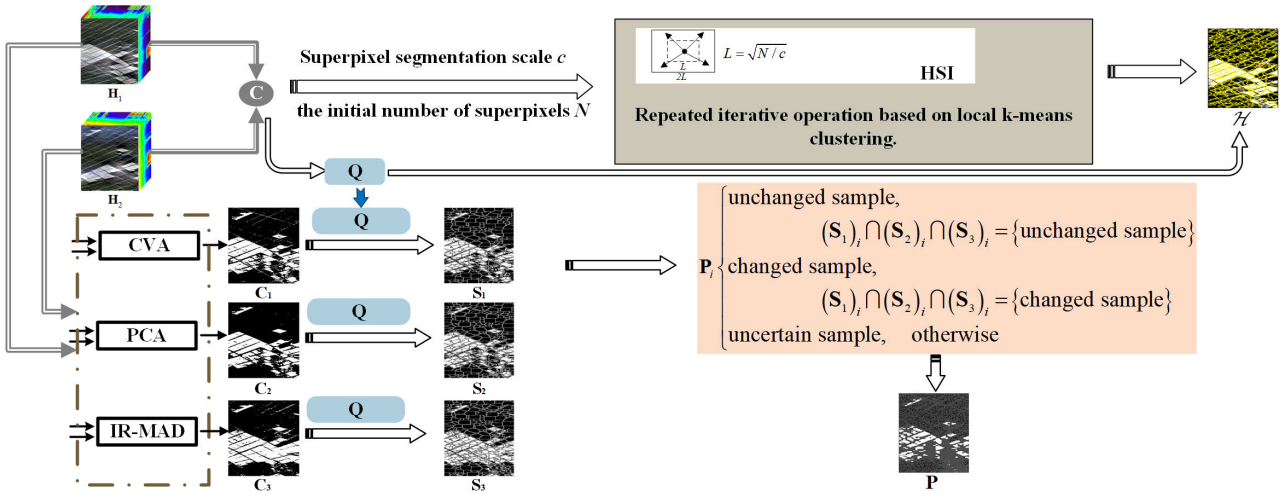


Fig. 2. Illustration of the superpixel-level joint decision-based high-confidence training samples selection.

are steep or contain privileged objects that make it impossible to label training samples. To solve this problem, in this article, a superpixel-level joint decision-based high-confidence training samples selection strategy is innovatively proposed, which fully exploits the spatial-spectral correlations between pixels in HSIs to provide high-confidence training sample labels for the unsupervised training of the network. The pixels contained in each superpixel region have high spatial-spectral similarity and can be approximately considered to be homogeneous. Therefore, we treat superpixel regions, rather than image patches or pixels, as the basic unit for training samples selection, which can effectively maintain the spatial-spectral correlation between pixels. Further, to improve the reliability of the training samples, in the joint decision-based selection strategy, the original CD maps obtained by the three traditional CD methods are comprehensively considered to obtain high-confidence pixels as the labels of training samples.

The specific process of superpixel-level joint decision-based high-confidence training samples selection is shown in Fig. 2. Given two input multi-temporal HSIs $\mathbf{H}_1 \in \mathbb{R}^{H\times W\times B}$ and $\mathbf{H}_2 \in \mathbb{R}^{H\times W\times B}$ that are captured at time t_1 and t_2 with the size of $H \times W$ and the number of spectral bands B, existing unsupervised CD methods including CVA, PCA, and IR-MAD are used to obtain the original CD maps $\mathbb{C} = \{\mathbf{C}_j \in \mathbb{R}^{H\times W}, j = 1, 2, 3\}$. For the purpose of obtaining pixels with high changed or unchanged probability, the three original CD maps are divided into a series of superpixel regions according to the superpixel distribution relationship of original multi-temporal HSI, in which the pixels in each superpixel region are likely to belong to the same image object. To obtain the distribution relationship of super-pixel regions, we use simple linear iterative clustering (SLIC) to perform superpixel segmentation on the original multi-temporal HSIs cascaded along the channel dimension. Meanwhile, the dimension of the cascaded image is reduced to 3 to adapt the input of SLIC algorithm. SLIC uses the local k-means clustering algorithm to generate compact and homogeneous superpixels, which effectively maintains the spatial-spectral correlation between pixels. The mapping relationship between the superpixel segmentation result $\mathcal{H} \in \mathbb{R}^{N\times B}$ and the original image can be represented as the association matrix $\mathbf{Q} \in \mathbb{R}^{HW\times N}$. Then association matrix $\mathbf{Q}$ is adopted to convert the initial CD maps $\mathbb{C}$ into the segmented CD maps $\mathbb{S}$ ($\mathbb{S} = \{\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3\}$) containing a series of superpixel regions. For each segmented CD map $\mathbf{S}_i$, the number of changed and unchanged pixels in each superpixel region is calculated respectively.

To ensure the reliability of the training samples, the segmented CD maps $\mathbb{S}$ obtained by the traditional unsupervised methods CVA, PCA, and IR-MAD are jointly used to make a decision to select the training sample. Considering the super-pixel regions at the same position in $\mathbf{S}_i$ obtained by these three methods, if all the pixels inside are changed, these pixels are determined to be labels of changed training samples. In the same way, if all pixels inside are unchanged, these pixels are considered as labels of unchanged training samples. Otherwise, the pixels are regarded as the labels of uncertain training samples. The specific superpixel-level joint-decision based rule can be described as follows:

$$\mathbf{P}_i = \begin{cases} \text{unchanged samples,} \\ \quad (\mathbf{S}_1)_i \cap (\mathbf{S}_2)_i \cap (\mathbf{S}_3)_i = \{\text{unchanged samples}\} \\ \text{changed samples,} \\ \quad (\mathbf{S}_1)_i \cap (\mathbf{S}_2)_i \cap (\mathbf{S}_3)_i = \{\text{changed samples}\} \\ \text{uncertain samples,} \\ \quad \text{otherwise} \end{cases} \tag{2}$$

where $\mathbf{P}_i$ denotes the labels of pixels in the ith superpixel region, $(\mathbf{S})_i$ denotes the set of pixel values in the ith superpixel region of the three segmented CD maps obtained by CVA, PCA, and IR-MAD methods, and $\cap$ is union operation. To address the imbalance of training samples, we further randomly select the same number of changed and unchanged training samples from the samples with certain labels.

According to the above rule, a large number of labeled pixels $p_x \in \mathbf{P}$ are selected as training sample labels. To extract sufficient and robust local features, the pixels in the $w \times w$ neighborhood of the pixel p_x in the original multi-temporal HSIs are selected as the training sample patch and its label

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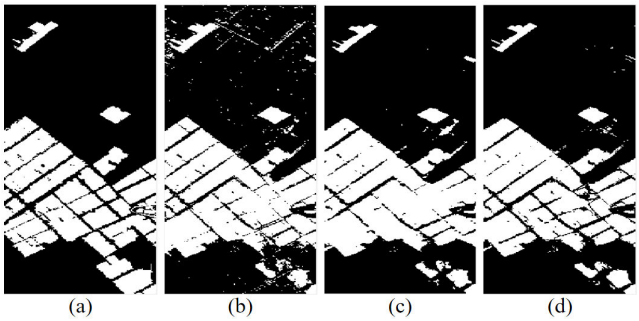


Fig. 3. CD results obtained from the proposed PCPTNet and its variants retaining only CPT subnetwork (variant-CPT) and only SPT subnetwork (variant-SPT) respectively on Farmland, River, and Santa Barbara datasets. (a) Ground-truth map. (b) Variant-CPT. (c) Variant-SPT. (d) PCPTNet.

is the central pixel p_x in the initial CD map. In this way, a large number of reliable training samples including changed and unchanged are collected to train the PCPTNet.

B. Parallel Compared-and-Stacked Pyramid Transformer Network

The main goal of the CD task is to compare the features of multi-temporal HSIs to identify the changes that occur on the objects in the images. Recently years, CNNs with excellent feature representation capabilities have been applied to many image processing tasks including CD. However, the CNN with a fixed convolution kernel size only has a limited receptive field, which cannot well explore the long-distance dependence between the spectral sequences. Fortunately, transformer, treating each spectral vector as an input sequence, that can effectively exploit the relationship between spectral sequences based on the self-attention mechanism is especially suitable for detecting changes in high-dimensional multi-temporal HSIs. In this article, a PCPTNet, which mainly includes CPT subnetwork, SPT subnetwork, and multilevel feature fusion and prediction network, is designed to distinguish the changed and unchanged classes in multi-temporal HSIs.

The CPT subnetwork can extract the difference features by comparing multiscale features obtained by transformer with the vector differencing comparison operator. However, the difference results are highly susceptible to external environmental conditions (e.g., light intensity and cloud cover) in multi-temporal HSIs and some subtle differences may be ignored. As shown in Fig. 3(b), the CD map obtained by a single CPT subnetwork contains a large number of white noise points and missed detection. The SPT subnetwork can effectively avoid the loss of image information by stacking the features obtained by the transformer, but it may be difficult to distinguish some tiny regions. As shown in Fig. 3(c), many unchanged pixels in the CD map obtained by SPT are wrongly identified as changed. To reduce the loss of information while obtaining the obvious difference information, a multilevel feature fusion and prediction network is proposed, in which the features at different scales obtained by the parallel CPT and SPT streams are fused through the squeeze-and-excitation networks (SENet) [40] and then sent to the classifier to obtain the final CD result. It can be clearly found from Fig. 3(d) that the missed and false detection in the CD map obtained by PCPTNet are significantly reduced.

1) CPT Subnetwork: The difference feature is the most critical information for identifying whether the corresponding pixels of the multi-temporal HSIs have changed. As the most direct way to obtain difference features, image comparison operations can effectively measure the distance and similarity of images by calculating the difference between pixel vectors. In addition, it is well known that difference features at different levels with different scales have unique advantages in maintaining detailed information and semantic information in multi-temporal HSIs, which will greatly facilitate the detection of changes. Inspired by these, the CPT subnetwork composed of compared feature transformer blocks (CFTBs) is proposed, in which a hierarchical pyramid structure is constructed to obtain sufficient multiscale difference features for CD.

Receiving the training sample patches $\mathbf{X}_1 \in \mathbb{R}^{w\times w\times B}$ and $\mathbf{X}_2 \in \mathbb{R}^{w\times w\times B}$ obtained in the unsupervised training sample selection stage as the inputs, the CPT subnetwork employs the transformer model to extract difference features of HSIs at different scales. The CPT subnetwork is constructed by stacking multilayer CFTBs, in which adjacent layers are connected by pooling layers. For CFTB, the transformer with strong ability of sequence modeling explores the relationship between the spectral sequences corresponding to each pixel in the input patch $\mathbf{X}_i \in \mathbb{R}^{w\times w\times B}$ in parallel. The transformer receives sequences as input, so we take the spectral vector corresponding to each pixel in the input image patch $\mathbf{X}_i \in \mathbb{R}^{w\times w\times B}$ as an input sequence. A series of spectral vectors are packed into a matrix $\mathbf{I}_i$ and fed into transformer together. In CPT subetwork, for the kth branch, the output of the lth CFTB represented as $\widehat{\mathbf{H}}_{C,k}^{(l)}$ is downsampling by max pooling as the input of the $(l+1)$th layer

$$\mathbf{H}_{C,k}^{(0)} = \mathbf{I}_i \mathbf{W}_{E(C,k)}^{(0)} + \mathbf{E}_{\text{pos}(C,k)}^{(0)} \tag{3-1}$$

$$\mathbf{H}_{C,k}^{(l+1)} = \text{pooling}\left(\widehat{\mathbf{H}}_{C,k}^{(l)}\right)\mathbf{W}_{E(C,k)}^{(l)} + \mathbf{E}_{\text{pos}(C,k)}^{(l)} \tag{3-2}$$

where $k = 1, 2$, $l = 0, 1, 2$, $\mathbf{W}_{E(C,k)}^{(l)}$ and $\mathbf{E}_{\text{pos}(C,k)}^{(l)}$ are the projection matrix and the position embedding of the kth branch of the lth layer CFTB, $\mathbf{H}_{C,k}^{(l)}$ and $\widehat{\mathbf{H}}_{C,k}^{(l)}$ respectively represent the input and output of the kth branch of the lth layer CFTB, and pooling$(\cdot)$ is the max pooling operation.

There exists a high correlation between the spectral sequences of pixels in an input patch, especially neighboring pixels that are close in spatial distance. The self-attention mechanism can effectively capture this dependency relationship between pixels by establishing a direct connection between pairs of pixels. The transformer adopts the MHSA mechanism to establish multiple self-attention subspaces focusing on different features of the objects, which is particularly effective for the recognition of objects in HSIs with complex scenes. Therefore, in the CPT subnetwork, the transformer is introduced to fully exploit the correlation between the center pixel and neighbor pixels in the input patch matrix of HSI. Then, in each CFTB, the difference feature is obtained by comparing the output features of the transformers in the two branches with differencing comparison operator.

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This procedure can be described as

$$\mathbf{Z}_{C,k}^{(l)} = \mathrm{MHSA}\left(\mathrm{LN}\left(\mathbf{H}_{C,k}^{(l)}\right)\right) + \mathbf{H}_{C,k}^{(l)} \tag{4-1}$$

$$\widehat{\mathbf{H}}_{C,k}^{(l)} = \mathrm{MLP}\left(\mathrm{LN}\left(\mathbf{Z}_{C,k}^{(l)}\right)\right) + \mathbf{Z}_{C,k}^{(l)} \tag{4-2}$$

$$\overline{\mathbf{H}}_{C}^{(l)} = \widehat{\mathbf{H}}_{C,1}^{(l)} - \widehat{\mathbf{H}}_{C,2}^{(l)} \tag{5}$$

where $k = 1, 2$, $l = 0, 1, 2$, MHSA$(\cdot)$, MLP$(\cdot)$, and LN$(\cdot)$ respectively represent the multihead self-attention, multilayer perceptron, and layer normalization operation, and $\overline{\mathbf{H}}_{C}^{(l)}$ represents the difference output of the lth layer CFTB. As the network deepens, the shallow detail information and deep semantic features that are essential for CD are sequentially extracted. In this way, the multilayer CFTBs are connected to form a feature pyramid network to obtain a multiscale difference feature representation. The sizes of the output features of the first, second and third comparison feature pyramid layers are 11×11, 5×5, and 3×3, respectively.

2) SPT Subnetwork: Generally, the fused paired features of HSIs can provide sufficient detail and semantic information for the CD task to help identify the objects in the HSIs. The existing feature fusion strategies in CD tasks are mainly divided into two categories: image comparison and image stacking. In CPT subnetwork, the image comparison strategy is adopted to obtain the multiscale difference feature of multitemporal HSIs. However, the comparison operation is very sensitive to noise and may result in the unreliable difference maps. To address this issue, an SPT subnetwork based on an image stacking strategy is proposed to obtain multiscale fused features for CD task.

The SPT applies a cascade of stacked feature transformer blocks (SFTBs) with a max pooling layer between them to extract the feature maps from the input multi-temporal HSIs. In each SFTB, the paired multi-temporal HSI patches are sent to the transformer for feature extraction. Then, to fuse the features of multi-temporal HSIs for CD, the output features of the transformers are stacked along the channel dimension to generate fused feature maps. In this way, each SFTB layer of the SPT can generate a fused feature map with a size of 11×11, 5×5, and 3×3, respectively. This process can be formulated as

$$\mathbf{Z}_{S,k}^{(l)} = \mathrm{MHSA}\left(\mathrm{LN}\left(\mathbf{H}_{S,k}^{(l)}\right)\right) + \mathbf{H}_{S,k}^{(l)} \tag{6-1}$$

$$\widehat{\mathbf{H}}_{S,k}^{(l)} = \mathrm{MLP}\left(\mathrm{LN}\left(\mathbf{Z}_{S,k}^{(l)}\right)\right) + \mathbf{Z}_{S,k}^{(l)} \tag{6-2}$$

$$\mathbf{H}_{S,k}^{(l+1)} = \mathrm{pooling}\left(\widehat{\mathbf{H}}_{S,k}^{(l)}\right)\mathbf{W}_{E(S,k)}^{(l)} + \mathbf{E}_{\mathrm{pos}(S,k)}^{(l)} \tag{6-3}$$

$$\overline{\mathbf{H}}_{S}^{(l)} = \mathrm{Concat}\left(\widehat{\mathbf{H}}_{S,1}^{(l)}, \widehat{\mathbf{H}}_{S,2}^{(l)}\right) \tag{7}$$

where $k = 1, 2$, $l = 0, 1, 2$, $\mathbf{H}_{S,k}^{(0)} = \mathbf{I}_i\mathbf{W}_{E(S,k)}^{(0)} + \mathbf{E}_{\mathrm{pos}(S,i)}^{(0)}$, $\mathbf{H}_{S,k}^{(l)}$ and $\widehat{\mathbf{H}}_{S,i}^{(l)}$ respectively represent the input and output of the kth branch of the lth layer SFTB, and $\overline{\mathbf{H}}_{S}^{(l)}$ represents the stacking output of the lth layer SFTB. The stacking operation can effectively reduce the loss of feature information of multi-temporal HSI and provide sufficient features for CD.

3) Multilevel Feature Fusion and Prediction Network: Combining the advantages of the CPT in directly obtaining different features and the advantages of the SPT in avoiding information loss, a multilevel feature fusion and prediction network is proposed, in which the SENet module is used to fuse the multiscale features obtained by the two streams and highlight useful features. The shallow features obtained in the feature pyramid subnetwork of the two streams contain a large amount of spatial detail information, and the deep features contain rich semantic information. The proposed multilevel feature fusion subnetwork can comprehensively utilize these multilevel features to improve the performance of CD.

The outputs $\overline{\mathbf{H}}_{C}^{(l)}$ and $\overline{\mathbf{H}}_{S}^{(l)}$ of the lth level of the CPT stream and the SPT stream are fused through a concatenation operation along the channel dimension and then sent to the SENet module to emphasize the useful channel features. The SENet module is mainly composed of squeeze, excitation, and reweight operations, where the squeeze operation shrinks the feature map on each channel to a real number with a global receptive field, the excitation operation models the correlation between feature channels by generating a weight for each feature channel, and the reweight operation multiplies the obtained channel weight by the corresponding channel feature to give attention to each channel feature. As a result, the fused feature $\widetilde{\mathbf{H}}^{(l)}$ of the lth layer output features of the CPT subnetwork and the SPT subnetwork is as follows:

$$\widetilde{\mathbf{H}}^{(l)} = f_{\mathrm{re}}\left(f_{\mathrm{ex}}\left(f_{\mathrm{sq}}\left(\mathrm{Concat}\left(\overline{\mathbf{H}}_{C}^{(l)}, \overline{\mathbf{H}}_{S}^{(l)}\right)\right)\right)\right) \tag{8}$$

where $f_{\mathrm{sq}}(\cdot)$, $f_{\mathrm{ex}}(\cdot)$, and $f_{\mathrm{re}}(\cdot)$ denote squeeze, excitation and reweight operations of SENet, respectively.

The fused features of the different layers of feature pyramids in the two parallel streams obtained by the SENet module can effectively maintain difference features and stacking features to reduce information loss and improve the robustness of the network. To make full use of the multilevel fused information, the outputs of the SENet module are further fused through max pooling and cascading operations as follows:

$$\widehat{\mathbf{H}}^{(l)} = \mathrm{Concat}\left(\mathrm{pooling}\left(\widetilde{\mathbf{H}}^{(l-1)}\right), \widetilde{\mathbf{H}}^{(l)}\right) \tag{9}$$

where $\widehat{\mathbf{H}}^{(l)}$ represents the fused feature of the output of lth layer SENet and the max-pooled output of $l-1$th layer SENet. Finally, the output features of the three layers of SENet are fused and sent to a fully connected classifier to generate a predicted CD map $\widehat{\mathbf{P}}$ as

$$\begin{aligned}\widehat{\mathbf{P}} &= f_F\left(\widehat{\mathbf{H}}^{(2)}\right) \\ &= f_F\Big(\mathrm{Concat}\Big(\mathrm{pooling}\Big(\mathrm{Concat}\Big(\mathrm{pooling}\Big(\widetilde{\mathbf{H}}^{(0)}\Big), \\ &\qquad \widetilde{\mathbf{H}}^{(1)}\Big)\Big), \widetilde{\mathbf{H}}^{(2)}\Big)\Big)\end{aligned} \tag{10}$$

where $f_F(\cdot)$ stands for fully connected layer. In this way, the multilevel feature fusion and prediction network uses the SENet channel attention module to achieve the fusion of multiscale features in the two parallel streams, which provides sufficient details and semantic information for detecting changes in multi-temporal HSIs.

In the entire learning process of the PCPTNet, the training samples obtained from the superpixel-level joint decision-based high-confidence training samples selection

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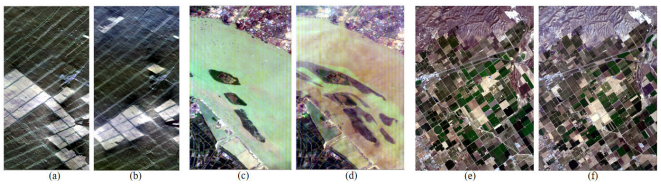


Fig. 4. Illustration of datasets. (a) Farmland image acquired on May 3, 2006. (b) Farmland image acquired on May 3, 2013. (c) River image acquired on May 3, 2006. (d) River image acquired on December 31, 2013. (e) Santa Barbara image acquired in 2013. (f) Santa Barbara image acquired in 2014.

stage are used for network training. The most commonly used cross-entropy loss function L_{ce} in CD tasks is adopted as follows:

$$L_{\text{ce}} = -\frac{1}{n}\sum_{e=1}^{n}\left(\widehat{\mathbf{P}}_e \log \mathbf{P}_e + \left(1-\widehat{\mathbf{P}}_e\right)\log(1-\mathbf{P}_e)\right) \quad (11)$$

where n represents the number of training samples, $\mathbf{P}$ is the labeled training samples selected by utilizing the traditional unsupervised CD methods, and $\widehat{\mathbf{P}}$ is the prediction results of the entire network. L_{ce} can effectively reflect the difference between the prediction results and the labels of the training samples. By continuously updating the parameters in the proposed PCPTNet, the value of the loss function L_{ce} is continuously reduced and the ability of the network to detect changes in the HSI is gradually improved.

IV. Experiment Results and Analysis

A. Datasets

The first dataset used in the experiments, named Farmland,[1] consists of two real multi-temporal HSIs collected from an agricultural area in Yancheng City, Jiangsu Province, China. As shown in Fig. 4(a) and (b), the HSIs were captured by the Hyperion sensor carried on Earth Observing-1 (EO-1) on May 3, 2006 and April 23, 2007, respectively. Each HSI contains 420 × 140 pixels with a spatial resolution of 30 m. This dataset covers 154 bands in the spectral range of 0.4–2.5 μm and the spectral resolution is 10 nm.

The second dataset, River,[2] was captured by the EO-1 Hyperion sensor on May 3, 2013 and December 31, 2013 [see Fig. 4(c) and (d)]. It covers a river area in Jiangsu Province, China. Each HSI contains 463 × 241 pixels and 198 bands with a spectral range of 0.4–2.5 μm. The spatial and spectral resolutions of each HSI are 30 m and 10 nm, respectively.

The third dataset is Santa Barbara,[3] which contains two registered HSIs obtained by the airborne visible infrared imaging spectrometer (AVIRIS) sensor over the city of Santa Barbara region, California. The multi-temporal HSIs were taken in 2013 [see Fig. 4(e)] and 2014 [see Fig. 4(f)]. Each HSI with a size of 984 × 740 contains 224 bands.

B. Evaluation Metrics and Implementation Details

To evaluate the performance of different CD methods, the accuracy evaluation indicators overall accuracy (OA), kappa coefficient (KC), precision, recall, and $F1$-score are adopted to measure the similarity between the predicted map and the ground-truth map. OA represents the ratio of the number of correctly classified pixels to the number of all pixels. KC compares the observed accuracy with the expected accuracy. Precision represents the proportion of the correctly predicted changed pixels to the total predicted changed pixels. Recall is the proportion of the correctly predicted changed pixels to the total changed pixels in ground-truth map. $F1$-score is the harmonic mean of precision and recall.

All the experiments in this article are implemented on the PyTorch platform, which is powered by a server equipped with the NVIDIA GTX 2080Ti GPU. We choose the adaptive moment estimation (Adam) optimizer with a learning rate set to 0.0001 for network training. The batch size is set to 16 and the number of training epoch is 600. The patch size is 11 × 11 and the numbers of the heads in transformer is 4. In addition, the superpixel segmentation scales on Farmland, River, and Santa Barbara datasets are set to 20, 10, and 50, respectively. The compactness value and maximum number of iterations are set to 1 and 20, respectively. The number of high-confidence training samples for China, USA, and Santa Barbara datasets are 14 862 (25%), 10 306 (9%), and 23 850 (3%), respectively. In sperpixel-level joint decision-based high-confidence training samples selection stage, for the CVA and IR-MAD methods, the k-means clustering method is used to generate the final CD map from the difference map. The number of iterations of the IR-MAD method is set as 30 to realize optimal detection performance of the network. The PCA method can achieve the best performance with the proportion of principal components corresponding to Farmland, River, and Santa Barbara datasets setting as 75%, 60%, and 75%, respectively.

C. Parameter Setting and Analysis

1) Parameter Settings of Superpixel Segmentation: In SLIC, local k-means clustering is performed iteratively to generate segmentation results. It can be seen that different segmentation scales, compactness values, and maximum numbers of iterations often lead to different distribution of superpixel regions, thus affecting the detection results. To further investigate the influence of different parameters on network performance, we conduct experiments on Farmland and River datasets to show the detection performance with different parameter settings. As can be seen from Fig. 5, for the Farmland and River datasets, the detection accuracy of the network is different under different parameter settings. According to the experimental results, we set the segmentation scales of Farmland, River, and Santa Barbara datasets to be 20, 10, and 50, respectively. The compactness value and maximum number of iterations on the three datasets are determined as 1 and 20, respectively.

2) Patch Size: In this article, to investigate the effect of the patch sizes on network performance, we conduct experiments under the input patch sizes of 5 × 5, 7 × 7, 9 × 9, 11 × 11, and 13 × 13. The detection results on Farmland and River datasets are shown in Fig. 6. It is easy to find that when the patch size is too large or too small, the performance of the net-

[1] https://rslab.ut.ac.ir/data
[2] https://pan.baidu.com/s/14ht8k5H-8ObzHJS6msYZjQ
[3] https://citius.usc.es/investigacion/datasets/hyperspectral-change-dataset

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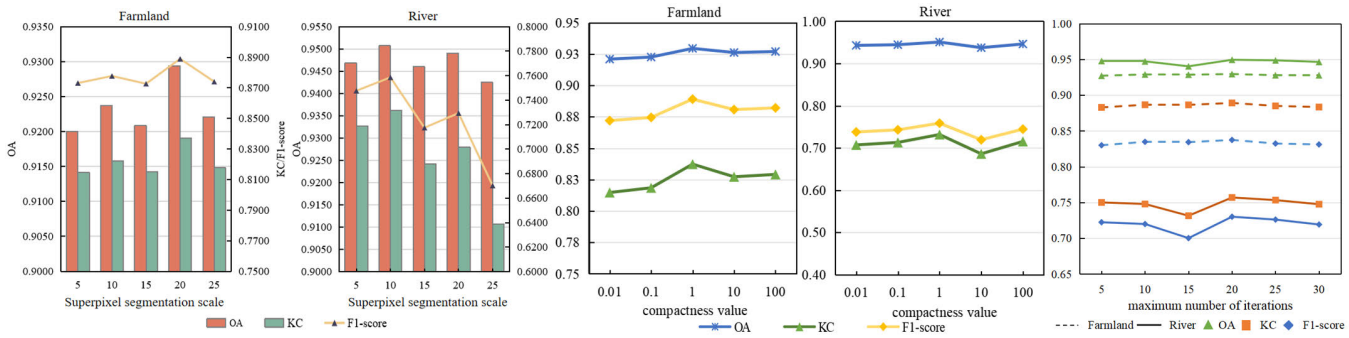


Fig. 5. CD results under different parameter settings in SLIC on Farmland and River datasets.

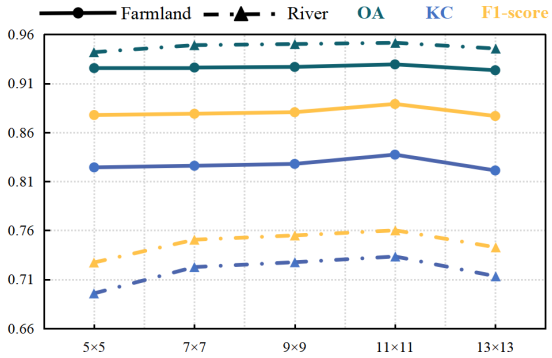


Fig. 6. CD results of PCPTNet with different patch sizes as input on Farmland and River datasets.

work will decline. This is because the information of a small patch is often insufficient. As the patch size increases, more information can be used to improve the detection performance. However, when the patch size is too large, multiple objects may be contained in the same patch, resulting in information redundancy and affecting detection accuracy. The experimental results show that the detection accuracy reaches the highest when the patch size is 11×11, so we set the patch size to 11×11 in the proposed PCPTNet.

3) Number of Training Samples: For deep learning models, the parameters of the network are continuously updated under the guidance of the training samples to learn the feature representation of the input data. Therefore, the number of training samples has a large impact on the automatic feature learning ability of the model. To analyze the influence of the number of training samples on the CD accuracy of the network, we conduct experiments with different numbers of training samples. Different percentages of samples are randomly selected from the original training samples as the final training samples, and the results are shown in Fig. 7. It can be found from the general trend that the detection accuracy is improved with the increase of the number of training samples. It is noteworthy that the network achieves well detection performance even with very few training samples, which strongly demonstrates the stability and superiority of the proposed PCPTNet. Finally, all the original high-confidence training samples obtained by superpixel-level joint decision-based training samples selection strategy are used for network training in PCPTNet, and the number of training samples for Farmland, River, and Santa Barbara datasets are 14 862, 10 306, and 23 850, respectively.

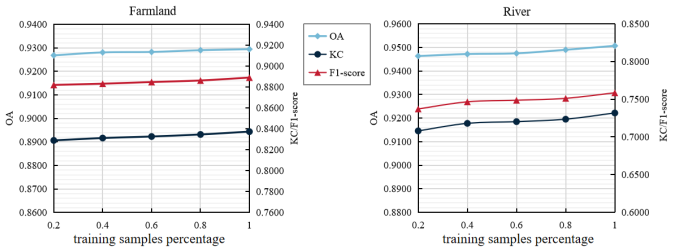


Fig. 7. CD results under different number of training samples on Farmland and River datasets.

D. Experiment Result and Analysis

To validate the effectiveness of the proposed method, nine advanced CD methods are adopted for comparative experiments, including IR-MAD [13], SVM [14], constrained nonnegative matrix factorization for hyperspectral change detection (CNMF) [41], tensor decomposition and reconstruction detector (TDRD) [42], bilinear convolutional neural networks (BCNNs) [43], recurrent convolutional neural network (ReCNN) [44], HyperNet [45], bitemporal image transformer (BIT) [46], and hyperspectral change detection via deep learning self-generated credible labels (SGCL) [30]. IR-MAD is a widely used traditional unsupervised CD method. SVM, as a typical classifier, processes image data based on statistical learning theory to detect changes. TDRD and CNMF are subspace-based and unmixing-based unsupervised CD methods, respectively. BCNNs and ReCNN are widely used deep learning-based CD methods. BIT is a transformer-based method. HyperNet and SGCL are the state-of-the-art unsupervised deep learning-based methods. The subjective experimental results of different methods on the three datasets are shown in Figs. 8–10, where the white pixels represent the changed regions and the black pixels represent the unchanged regions. For the Santa Barbara dataset, gray pixels represent unlabeled regions. Meanwhile, since it is impossible to determine the category of gray pixels to evaluate the accuracy of detection results, gray samples are not selected as training samples. The objective results are list in Table I. To evaluate the deep learning-based methods objectively, each deep learning-based method runs five times on each dataset, with its average value and standard deviation as the measurement standard.

1) *Results for Farmland dataset:* The results of different methods on the Farmland dataset are shown in Fig. 8. It can be found that, in the CD maps obtained by SVM,

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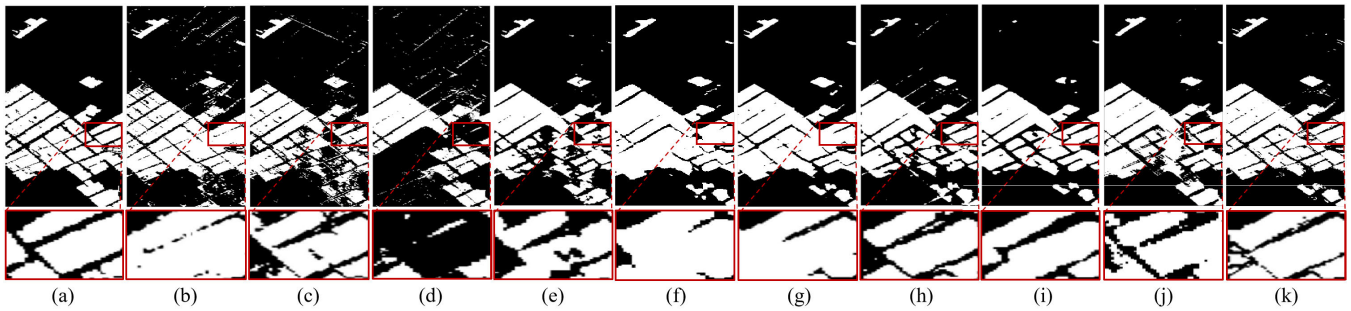


Fig. 8. CD results on Farmland dataset. (a) Ground-truth map. (b) IR-MAD. (c) SVM. (d) CNMF. (e) TDRD. (f) BCNNs. (g) ReCNN. (h) HyperNet. (i) BIT. (j) SGCL. (k) PCPTNet.

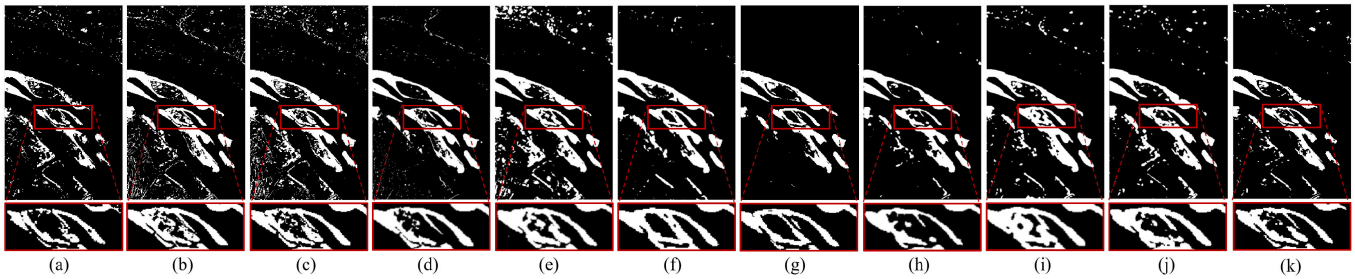


Fig. 9. CD results on River dataset. (a) Ground-truth map. (b) IR-MAD. (c) SVM. (d) CNMF. (e) TDRD. (f) BCNNs. (g) ReCNN. (h) HyperNet. (i) BIT. (j) SGCL. (k) PCPTNet.

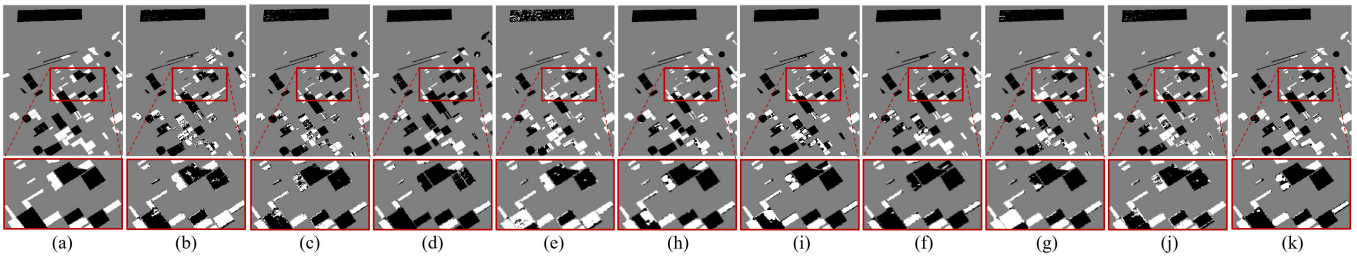


Fig. 10. CD results on Santa Barbara dataset. (a) Ground-truth map. (b) IR-MAD. (c) SVM. (d) CNMF. (e) TDRD. (f) BCNNs. (g) ReCNN. (h) HyperNet. (i) BIT. (j) SGCL. (k) PCPTNet.

TABLE I

QUALITY METRICS OBTAINED BY DIFFERENT HSI-CD METHODS ON THREE DATASETS

Method	IR-MAD	SVM	CNMF	TDRD	BCNNs	ReCNN	HyperNet	BIT	SGCL	PCPTNet
Farmland										
OA	0.8974	0.8880	0.8230	0.8873	0.9132±0.0009	0.9203±0.0021	0.9026±0.0005	0.9046±0.0111	0.9126±0.0036	**0.9294±0.0005**
KC	0.7632	0.7329	0.5546	0.7315	0.8019±0.0018	0.8163±0.0034	0.7681±0.0011	0.7831±0.0218	0.7954±0.0098	**0.8373±0.0013**
Precision	0.8267	0.8527	0.7994	0.8494	0.8378±0.0029	0.8610±0.0161	0.8758±0.0011	0.8297±0.0681	0.8672±0.0024	**0.8740±0.0051**
Recall	0.8501	0.7758	0.5790	0.7773	0.8960±0.0011	0.8892±0.0156	0.8023±0.0006	0.8894±0.0844	0.8306±0.0568	**0.9044±0.0065**
F1-score	0.8382	0.8125	0.6716	0.8117	0.8659±0.0010	0.8747±0.0017	0.8374±0.0008	0.8533±0.0156	0.8587±0.0075	**0.8890±0.0010**
River										
OA	0.9383	0.9437	0.9283	0.9220	0.9358±0.0007	0.9399±0.0037	0.9410±0.0024	0.9364±0.0044	0.9419±0.0001	**0.9507±0.0013**
KC	0.6798	0.7162	0.5890	0.6365	0.6438±0.0030	0.6563±0.0035	0.6564±0.0164	0.6591±0.0140	0.6845±0.0107	**0.7318±0.0062**
Precision	0.5983	0.6138	0.5717	0.5288	0.6008±0.0038	0.6290±0.0345	0.6396±0.0341	0.5979±0.0246	0.6228±0.0072	**0.6606±0.0066**
Recall	0.8823	**0.9516**	0.6972	0.9403	0.7799±0.0014	0.7673±0.0556	0.7556±0.0931	0.8270±0.0132	0.8437±0.0396	0.8907±0.0043
F1-score	0.7131	0.7462	0.6282	0.6769	0.6788±0.0027	0.6891±0.0041	0.6886±0.0174	0.6936±0.0118	0.7160±0.0105	**0.7586±0.0055**
Santa Barbara										
OA	0.8630	0.8793	0.8336	0.8600	0.9122±0.0115	0.9011±0.0094	0.9088±0.0160	0.8819±0.0079	0.9103±0.0021	**0.9426±0.0037**
KC	0.7099	0.7418	0.6297	0.7184	0.8144±0.0233	0.7816±0.0132	0.8044±0.0352	0.7546±0.0161	0.8131±0.0044	**0.8789±0.0084**
Precision	0.8464	0.8918	0.9318	0.7600	0.9144±0.0026	0.8973±0.0051	0.9414±0.0105	0.8347±0.0158	0.9354±0.0023	**0.9433±0.0102**
Recall	0.7962	0.7887	0.6225	0.9413	0.9435±0.0177	0.9376±0.0077	0.8191±0.0346	0.8730±0.0133	0.9153±0.0011	**0.9634±0.0061**
F1-score	0.8205	0.8371	0.7463	0.8410	0.9286±0.0100	0.9170±0.0052	0.8758±0.0234	0.8533±0.0093	0.9253±0.0017	**0.9532±0.0027**

CNMF, TDRD, HyperNet, and SGCL methods, some changed areas are erroneously ignored. For BCNNs and ReCNN methods, some subtle unchanged regions are wrongly classified as changed. Obviously, the proposed

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PCPTNet [see Fig. 8(k)] has the lowest error rate in CD compared with other methods. It can be clearly found from Table I that the detection accuracy of the traditional CD methods is relatively low. The performance of deep learning-based methods has been significantly improved. The most notable thing is that PCPTNet shows the most outstanding performance. In a word, both subjective and objective results have demonstrated the effectiveness of the proposed PCPTNet.

2) *Results for River dataset:* Fig. 9 visualizes the CD results obtained by different methods on the River dataset. It is evident that IR-MAD, SVM, TDRD, BIT, and SGCL methods incorrectly classify a large number of unchanged regions as changed. In addition, it is a pity that the CNMF, and HyperNet methods have difficulty in detecting subtle and scattered changes in River dataset. Fig. 9(k) strongly confirms the excellent performance of the proposed PCPTNet, in which almost most of the changed and unchanged regions can be successfully detected. In terms of objective results, although the recall of the traditional methods are relatively high, their precision is relatively lower than deep learning-based methods. Considering all the indicators comprehensively, the proposed PCPTNet achieves the best detection results.

3) *Results for Santa Barbara dataset:* The experiment results on Santa Barbara dataset are shown in Fig. 10. In CD maps obtained by traditional methods, a large number of changed and unchanged pixels cannot be distinguished correctly. As shown in Fig. 10(f)–(j), deep learning-based methods have a stronger ability to identify changes in multi-temporal HSIs than traditional methods. As we can see from Fig. 10(k), the proposed PCPTNet has the lowest error rate in CD compared with other methods. In addition, the objective results show that the accuracy of the results of the traditional methods are expected to be improved to an acceptable level. Deep learning-based methods can obtain improved detection accuracy. It can be observed that PCPTNet with strong feature modeling ability achieves the best performance.

E. Ablation Studies

1) Effect of Transformer Model: In this article, the transformer with the advantages of sequence modeling is introduced to the HSI-CD task. Compared with the CNN with limited receptive field, the transformer model uses the MHSA mechanism to fully mine the context relationship between pixels in the input image patch to determine the category of the center pixel, which is especially helpful for distinguishing objects in multi-temporal HSIs. To verify the effectiveness of the transformer model, we replace the transformer in the proposed PCPTNet with a 3×3 convolutional layer. The CD results obtained by PCPTNet based on transformer and its variant based on CNN (variant-CNN) on the three datasets are shown in Fig. 11. It can be clearly found that the performance of the proposed PCPTNet significantly outperforms its variant, which clearly shows that transformer model is beneficial to improve the detection accuracy.

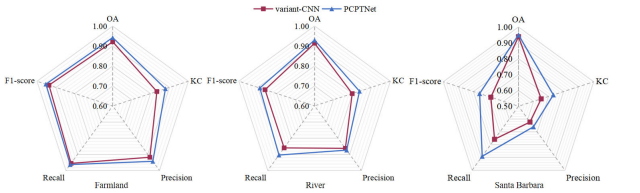


Fig. 11. CD results obtained by the proposed PCPTNet and its variant that replacing transformer with a convolutional layer (variant-CNN) on Farmland, River, and Santa Barbara datasets.

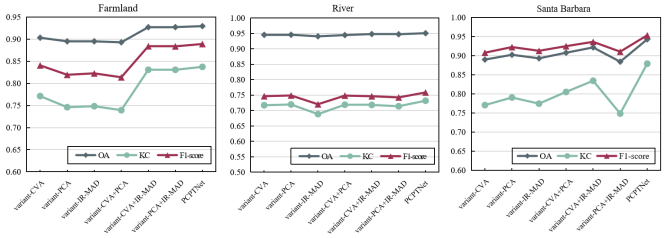


Fig. 12. CD results obtained by the proposed PCPTNet and its variants that using single decision (variant-CVA, variant-PCA, and variant-IR-MAD) and double decision (variant-CVA + PCA, variant-CVA + IR-MAD, and variant-PCA + IR-MAD) based training sample selection strategies on Farmland, River, and Santa Barbara datasets.

TABLE II
QUALITY METRICS OBTAINED FROM THE PROPOSED PCPTNET AND ITS VARIANTS RETAINING ONLY CPT SUBNETWORK (VARIANT-CPT) AND ONLY SPT SUBNETWORK (VARIANT-SPT) RESPECTIVELY ON FARMLAND, RIVER, AND SANTA BARBARA DATASETS

Datasets	Method	OA	KC	Precision	Recall	F1-score
Farmland	variant-CPT	0.9109	0.7976	0.8280	0.9024	0.8636
	variant-SPT	0.9220	0.8211	0.8553	0.9031	0.8786
	PCPTNet	**0.9294**	**0.8373**	**0.8740**	**0.9044**	**0.8890**
River	variant-CPT	0.9417	0.6893	0.6176	0.8656	0.7208
	variant-SPT	0.9341	0.6414	0.5900	0.7946	0.6772
	PCPTNet	**0.9507**	**0.7318**	**0.6606**	**0.8907**	**0.7586**
Santa Barbara	variant-CPT	0.9020	0.7990	**0.9560**	0.8789	0.9158
	variant-SPT	0.8452	0.6764	0.8756	0.8682	0.8719
	PCPTNet	**0.9426**	**0.8789**	0.9433	**0.9634**	**0.9532**

2) Effect of Joint Decision-Based Training Samples Selection Strategy: For an unsupervised CD method, it is crucial to select reliable training samples. In this article, a joint decision-based training samples selection strategy is designed to select high-confidence training samples, which jointly uses the results of three traditional unsupervised CD methods (CVA, PCA, and IR-MAD) to improve the reliability of training samples. The pixels are considered reliable only when the corresponding categories of the pixels at the same super-pixel regions in the CD maps obtained by the above three methods are consistent. To verify the robustness and effectiveness of the proposed joint decision-based training samples selection strategy, the detection results obtained by the proposed PCPTNet are compared with its variants that using single decision (variant-CVA, variant-PCA, and variant-IR-MAD) and double decision (variant-CVA+PCA, variant-CVA+IR-MAD, and variant-PCA+IR-MAD) based training sample selection strategies. As shown in Fig. 12, the proposed joint decision-based training samples selection strategy is more efficient than both single and double decision-based training

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TABLE III
COMPUTATIONAL COST OF DIFFERENT METHODS ON FARMLAND DATASET

Method	IR-MAD	SVM	CNMF	TDRD	BCNNs	ReCNN	HyperNet	BIT	SGCL	PCPTNet
Params	-	-	-	-	695.3k	3648.1k	282.5k	3574.0k	3092.5k	1782.2k
Running Time(s)	1.24	4.48	86.84	7.50	5.98	5.53	0.71	16.15	7.78	12.97

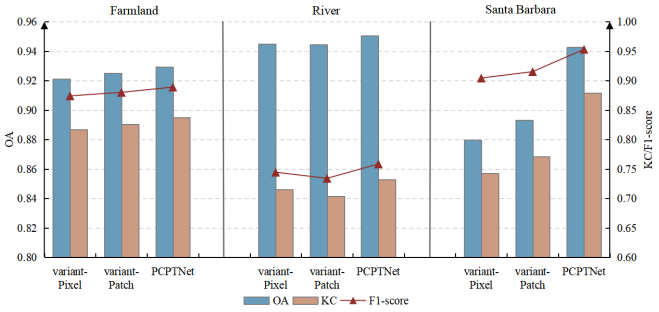


Fig. 13. CD results obtained by the proposed PCPTNet and its variants that using pixel level (variant-pixel) and patch level (variant-patch) joint decision-based training sample selection strategies on Farmland, River, and Santa Barbara datasets.

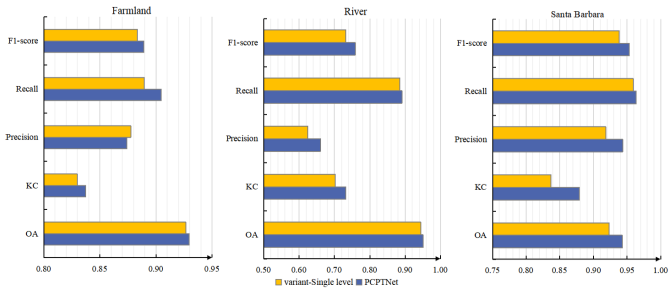


Fig. 14. CD results obtained by the proposed PCPTNet and its variant that replacing multilevel feature fusion with single-level feature fusion (variant-single level) on Farmland, River, and Santa Barbara datasets.

samples selection strategies, yielding the highest accuracy of CD results.

3) Effect of Superpixel Segmentation: In this article, superpixel regions, rather than image patches or pixels, are regarded as a basic unit for selecting training samples, which can not only make full use of the spatial-spectral correlation between pixels but also preserve the category boundary information to improve the reliability of the training samples. To validate the effectiveness of superpixel segmentation for training sample selection, we compare the CD results obtained by the proposed PCPTNet and its variants that using pixel level (variant-pixel) and patch level (variant-patch) joint decision-based training sample selection strategies on all the three datasets. From Fig. 13, the proposed PCPTNet using the superpixel level training sample selection strategy achieves the highest CD accuracy on the three datasets, which proves that the superpixel segmentation operation is effective to improve the detection performance of the network.

4) Effect of Parallel Compared-and-Stacked Pyramid Structure: In the proposed PCPTNet, the network structure has a great influence on the accuracy of the CD results. To make full use of the difference information of multi-temporal HSIs and avoid the loss of feature information, a parallel compared-and-stacked pyramid structure composed of a feature comparison network (i.e., CPT subnetwork) and a feature stacking network (i.e., SPT subnetwork) is designed for CD. To prove the superiority of the parallel compared-and-stacked pyramid structure, the CD results obtained from the proposed PCPTNet are compared with its variants with only CPT subnetwork (variant-CPT) and only SPT subnetwork (variant-SPT) respectively on the three datasets. It can be found from Fig. 3. that the result of the variant with CPT subnetwork that is sensitive to noise alone has a large number of missed detections and false detections, and the result of the variant with CPT subnetwork that can avoid information loss alone has serious false detections. Obviously, the proposed PCPTNet with the parallel compared-and-stacked pyramid structure has achieved the most outstanding results. The quantitative results are shown in Table II, from which it can be easily found that PCPTNet with parallel compared-and-stacked pyramid structure achieves the best CD performance when all metrics are considered comprehensively. In conclusion, both the visual and quantitative results demonstrate that the presented parallel compared-and-stacked pyramid structure has a strong ability in feature expression.

5) Effect of Multilevel Feature Fusion: In the proposed PCPTNet, a multilevel feature fusion network is used to comprehensively utilize the multiscale comparison features and stacking features generated by CPT subnetwork and SPT subnetwork. The fusion of multiscale feature maps can make full use of the low-level detailed features and high-level semantic features to improve the ability to discriminate objects in the multi-temporal HSIs. This point of view is strongly confirmed by the experimental results shown in Fig. 14, which compares the accuracy of the CD results generated by the proposed PCPTNet and its variant that replacing multilevel feature fusion with only single level (i.e., the last layer of the feature pyramid structure) feature fusion (variant-single level) on the three datasets. It can be found that the performance of our proposed PCPTNet is better than that of its variant, which strongly demonstrates the effectiveness of multilevel feature fusion to improve the detection accuracy of the network.

6) Computational Efficiency: The computational cost of different methods including running time and the number of parameters on the Farmland dataset is recorded in Table III. All traditional methods are implemented in MATLAB (R2021b) on Intel[4] Core[5] i5-7200U CPU at 2.50 GHz 2.71 GHz, and deep learning-based methods are implemented on NVIDIA GeForce GTX 2080Ti GPU. It can be observed from Table III

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that the computational cost of traditional methods is satisfactory, but the CD performance is significantly inferior to the deep learning related methods (i.e., ReCNN, BCNNs, HyperNet, and BIT) and the proposed PCPTNet. Compared with traditional methods, most deep learning-based methods require more time to train the model, but the performance is improved. Especially PCPTNet obtains the best CD performance. In terms of parameter quantity, PCPTNet achieves a balance between performance and parameter quantity. In future work, we will focus on developing more efficient models to reduce computational cost while maintaining model performance in CD tasks.

V. Conclusion

In this article, we propose a novel unsupervised method named PCPTNet for unsupervised HSI-CD. In the unsupervised framework, a superpixel-level joint decision-based high-confidence training samples selection strategy is presented to obtain reliable training samples. The proposed PCPTNet mainly builds a parallel compared-and-stacked pyramid structure based on the transfomer model with the ability to capture the context information of input image patches, which acquires multiscale difference features and stacking features of multi-temporal HSIs. Our experiments show that fusing difference features obtained by CPT and stacking features obtained by SPT can reduce false and missed detections, thereby improving detection accuracy. For future work, it is very interesting to extend this model to noisy environments to improve the generalization and robustness.

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